He found, not really surprisingly, that the exchange rate is different, depending upon how starved a bat is. A given amount of blood adds more hours to the life of a highly starved bat than to a less starved one. In other words, although the act of donating blood would increase the chances of the donor dying, this increase was small compared with the increase in the recipient's chances of surviving. Economically speaking, then, it seems plausible that vampire economics conform to the rules of a Prisoner's Dilemma. The blood that the donor gives up is less precious to her (social groups in vampires are female groups) than the same quantity of blood is to the recipient. On her unlucky nights she really would benefit enormously from a gift of blood. But on her lucky nights she would benefit slightly, if she could get away with it, from defecting—refusing to donate blood. 'Getting away with it', of course, means something only if the bats are adopting some kind of Tit for Tat strategy. So, are the other conditions for the evolution of Tit for Tat reciprocation met?

In particular, can these bats recognize one another as individuals? Wilkinson did an experiment with captive bats, proving that they can. The basic idea was to take one bat away for a night and starve it while the others were all fed. The unfortunate starved bat was then returned to the roost, and Wilkinson watched to see who, if anyone, gave it food. The experiment was repeated many times, with the bats taking turns to be the starved victim. The key point was that this population of captive bats was a mixture of two separate groups, taken from caves many miles apart. If vampires are capable of recognizing their friends, the experimentally starved bat should turn out to be fed only by those from its own original cave.

That is pretty much what happened. Thirteen cases of donation were observed. In twelve out of these thirteen, the donor bat was an 'old friend' of the starved victim, taken from the same cave; in only one out of the thirteen cases was the starved victim fed by a 'new friend', not taken from the same cave. Of course this could be a coincidence but we can calculate the odds against this. They come to less than 1 in 500. It is pretty safe to conclude that the bats really were biased in favour of feeding old friends rather than strangers from a different cave.

Vampires are great mythmakers. To devotees of Victorian Gothic they are dark forces that terrorize by night, sapping vital fluids, sacrificing an innocent life merely to gratify a thirst. Combine this with that other